

CSI 3140 — WWW Structures, Techniques and Standards

Laboratory 2 — CSS Styling, Box Model, Responsive Design, and Flexbox/Grid Layout

Professor: Mohamed Ali Ibrahim, PhD, PEng **Term:** Spring/Summer 2026

Submission: Brightspace —

Team Members:

Name	Student Number
[David]	[Student #300308910]
[Neelmani Bhardwaj]	[Student #300389998]
[Liam]	[Student #300356748]

1. Website Description

Our website is a fictional fitness club called **Uranus Fitness**; a child company of Planet Fitness offering different plans to sign up. The site introduces the club, describes its programs and workshop structure, presents a membership-plan schedule with pricing in a structured table, and lets visitors register for a membership tier through an accessible HTML form. The build employs standards-based HTML5 and accessibility-aware markup; CSS is used.

2. File Structure

Lab2/

— index.html	– Home page introducing Uranus Fitness (hero image + nav)
— about.html	– Club description: headings, lists, figure, external + email links
— schedule.html	– Membership-plan schedule as a captioned table
— registration.html	– Accessible registration form
— styles.css	– Basic CSS for layout, banner, readability
— img/	
— logo.png	– Banner logo, used in header on all pages
— old_guypng.png	– Hero image on the home page

3. CSS Organization and Design Choices

CSS organization

The stylesheet is divided into clear sections: a base/reset area, the shared header/navigation, hero/intro rules, panel/card components, form styles, table/schedule styles, and responsive helpers. Each section groups related selectors and layout rules so new components can be added beneath the matching comment block. A single media query at max-width: 800px collapses multi-column grids to a single column and adjusts navigation behavior for small screens.

Major selectors and classes

Key layout selectors include `.site-nav` (flex navigation), `.hero` (two-column grid for intro + visual), `.feature-grid` and `.about-grid` (card grids), and `.form-grid` (two-column form layout). Reusable elements include `.brand/.banner-logo` for identity, `.nav-links` for navigation links, `.button` variants for actions, `.feature-card/.feature-panel` for card content, `.registration-form` and `.form-fieldset` for forms, and `.schedule-table` along with `.table-wrap` for responsive tables. The small label class `.eyebrow` is used consistently above section headings.

Visual design choices

Typography: the site uses the system UI stack (system-ui, sans-serif) for consistent native rendering. Headings in the hero area use `clamp()` to scale responsively while keeping a clear visual hierarchy. Small labels (the `.eyebrow`) use uppercase and increased letter-spacing to sit visually above headings without competing with them.

Colours: the palette centers on a light blue background gradient for an airy tone, with a deep blue (#10212b) as the main brand/app color used for primary buttons and table headers. A lighter teal (#7dd3fc) is used as an accent for subtle highlights (nav underline, accents), and cards use soft white/pale-blue fills to lift content off the page.

Spacing & rhythm: spacing follows a comfortable 1rem–2rem scale for padding and gaps. The hero grid uses a 2rem gap, while card and form grids use 1rem gaps to maintain readability. Border radii are modestly rounded (roughly 0.8rem–1.5rem) to unify the look of images, panels, and inputs.

4. CSS Box Model, Flexbox, and Grid

CSS Box Model:-

The website uses the CSS model to control spacing, sizing and visual separation between sections. At the start of the stylesheet we used `: * { margin: 0; padding: 0; box-sizing: border-box; }`. This makes the layout easier to control because padding and borders are included inside the total width of an element. Without this rule, cards, tables, images and form fields could become wider than expected and can cause overflow issues.

The main page content uses a centred container with minimum width. This keeps the website readable on large screens and prevents text lines from becoming too long. Padding

is used inside cards, form fieldsets, buttons, table cells and panels to create comfortable spacing. Margins and gaps are used between sections so the content does not look crowded.

Borders. Border-radius and box shadows are also used to separate content visually. For example, `.feature-panel`, `.feature-card`, `.form-fieldset`, and `.hero-visual` use padding, rounded corners, and light shadows to make each section look like a clear content block.

The box model issue we checked was horizontal overflow on small screens, especially on the schedule table. The table was placed inside `.table-wrap` with responsive handling so it does not break the page layout on mobile.

Flexbox:-

Flexbox is used for one-dimensional layouts where items need to be aligned in a row or wrapped neatly.

The main navigation uses Flexbox through the `.site-nav` class. It uses `display:flex`, `justify-content: space-between`, `align-items:center`, `gap` and `flex wrap`. This lets the logo and navigation links stay aligned on desktop while still wrapping properly on smaller screens,

Flexbox is also used in `.nav-links`, `.hero-actions` and `.form-actions`. These sections contain links or buttons that need clean horizontal spacing. Flexbox is appropriate here because these elements mainly need alignment in one direction, either across a row or wrapped around when the screen is smaller.

Grid:

CSS Grid is used for large two-dimensional layouts where content needs rows and columns.

The `.hero` section uses a grid to create two columns for the title and text, and one column for the image. This gives the home , about, schedule and registration a balanced desktop layout.

The `.feature-grid` class is used to create a three-column card layout. The `.about-grid` class creates a two-column layout for larger content sections, the `.form-grid` class creates a two-column form layout for fields such as name, email and phone number.

5. Responsive Design

The website was designed to work on desktop, tablet and mobile screen sizes. Each HTML page includes the view port meta tag:<meta name="viewport" content="width=device-width, initial-scale=1.0">. This is important because it tells the browser to use the real width of the device instead of sharing the layout onto a mobile screen.

The design used flexible CSS units such as `rem`, `%`, `fr`, `min()`, `calc()`, and `clamp()`. These units help the layout resize naturally instead of depending on fixed pixel value.

The main responsive breakpoint is @media (max-width: 800px)

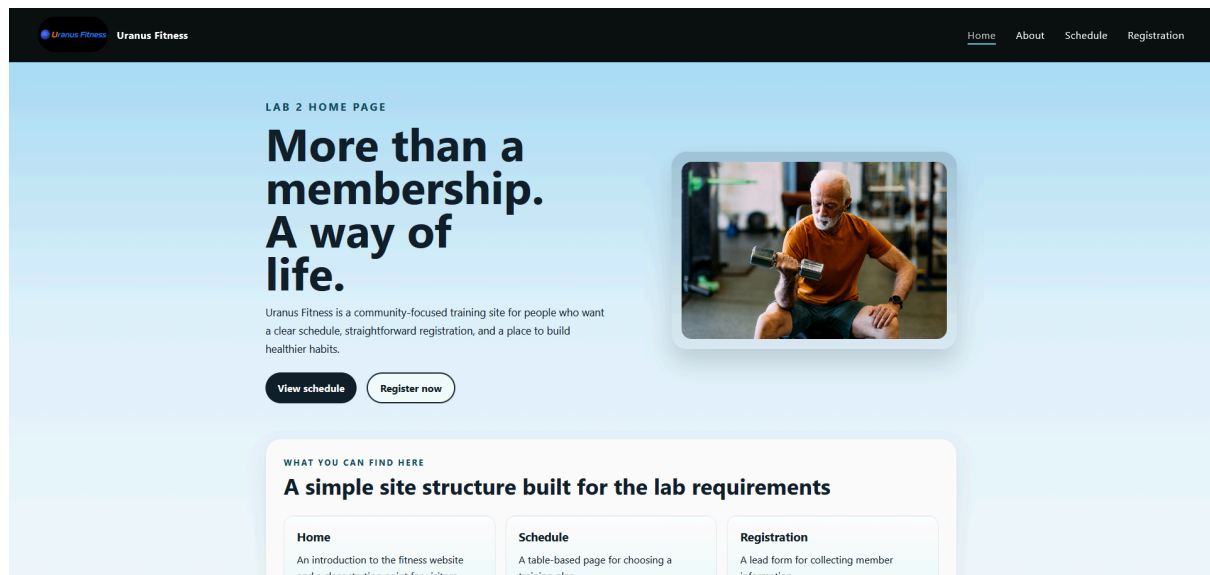
At this screen width, the layout changes to better fit smaller devices. The navigation becomes centered and wrapped if needed. Most of the section changes from multiple to single column. This prevents the content from getting too squeezed on mobile phones.

Images are responsive too because they use `max-width: 100%` and `display: block`. This keeps the images inside the container and prevents them from overflowing on the page.

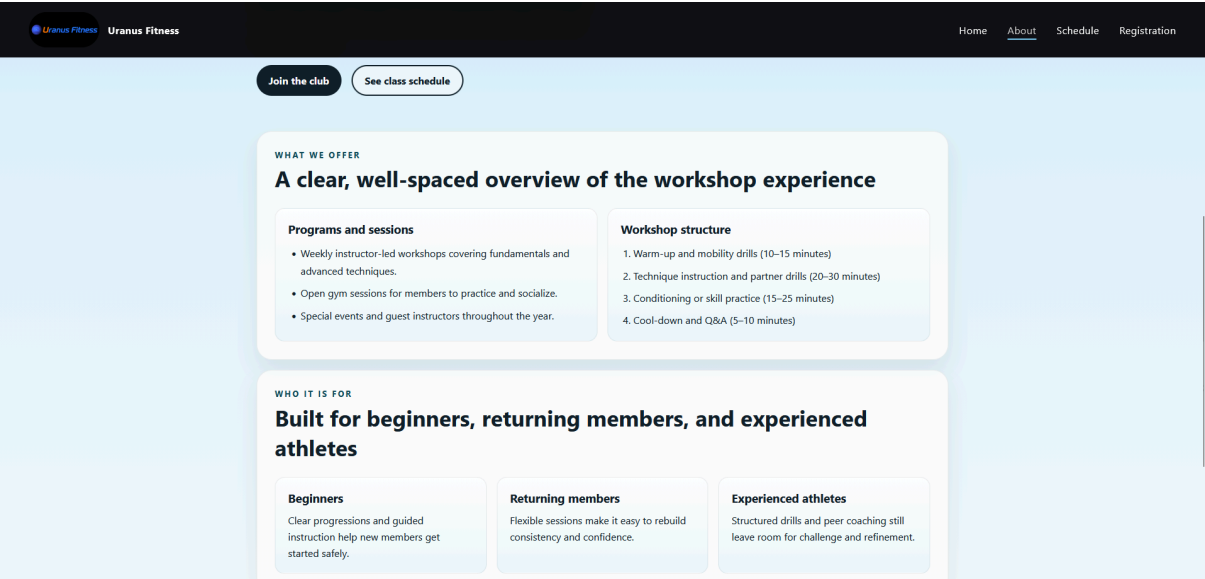
During testing the website was checked on desktop, tablet and mobile views, the layout stayed readable, navigation links remained readable and stable.

6. Screenshots

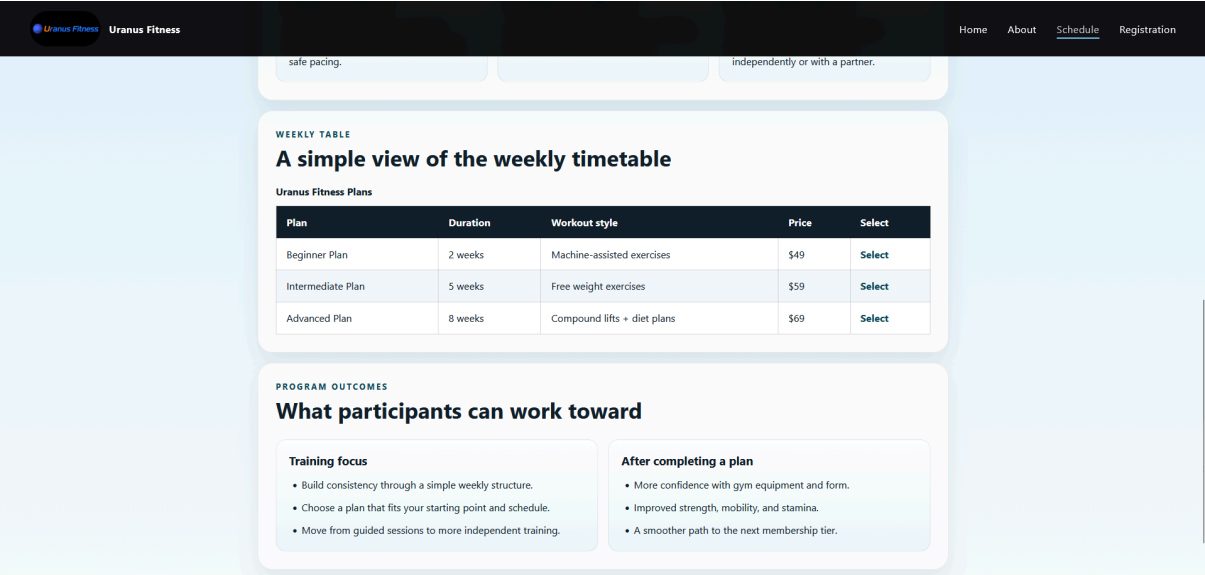
Desktop



Index.html



About.html



Schedule.html

Uranus Fitness

Uranus Fitness

HomeAboutScheduleRegistration

CONTACT FORM

Registration details

Personal information

Full name

Email address

Phone number

Age range

Membership selection

Choose a plan

☐ Beginner

☐ Intermediate

☐ Advanced

Preferred contact method

Preferred start date

About you

Tell us about your goals

What do you want to focus on?

registration.html

Tablet

Uranus Fitness

Uranus Fitness

HomeAboutScheduleRegistration

LAB 2 HOME PAGE

More than a membership.
A way of life.

Uranus Fitness is a community-focused training site for people who want a clear schedule, straightforward registration, and a place to build healthier habits.

View schedule

Register now

WHAT YOU CAN FIND HERE

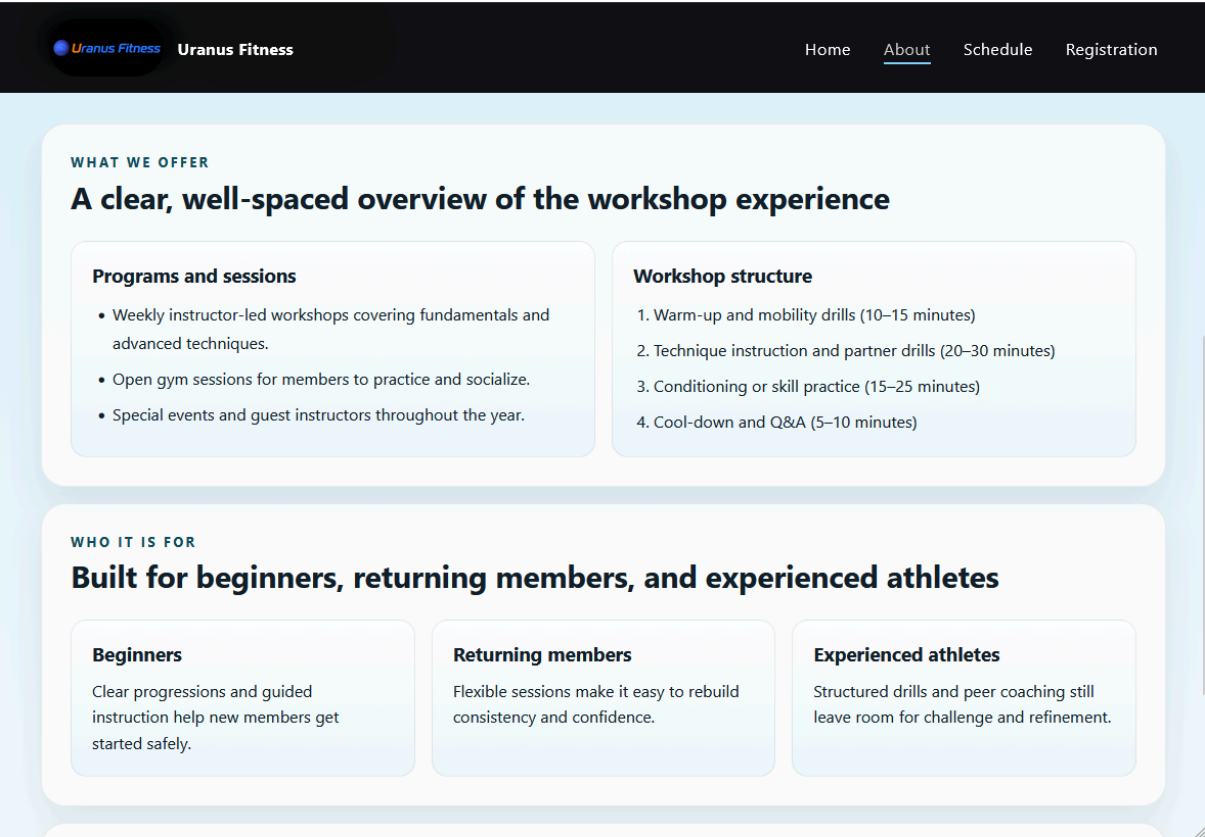
A simple site structure built for the lab requirements

Home

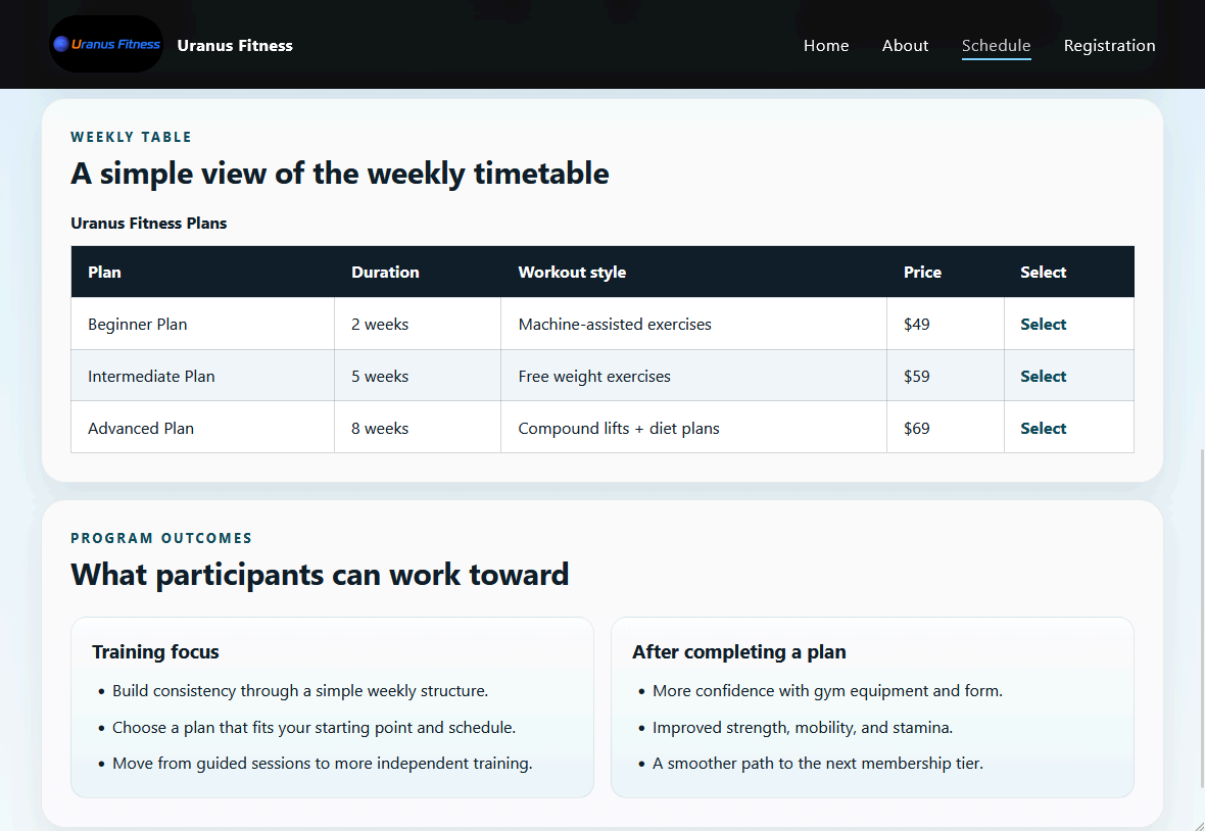
Schedule

Registration

Index.html



About.html



Schedule.html

any screen size.

CONTACT FORM

Registration details

Personal information

Full name

Email address

Phone number

Age range

Select an option



Membership selection

Choose a plan

- ☐ Beginner
- ☐ Intermediate
- ☐ Advanced

Preferred contact method

Email

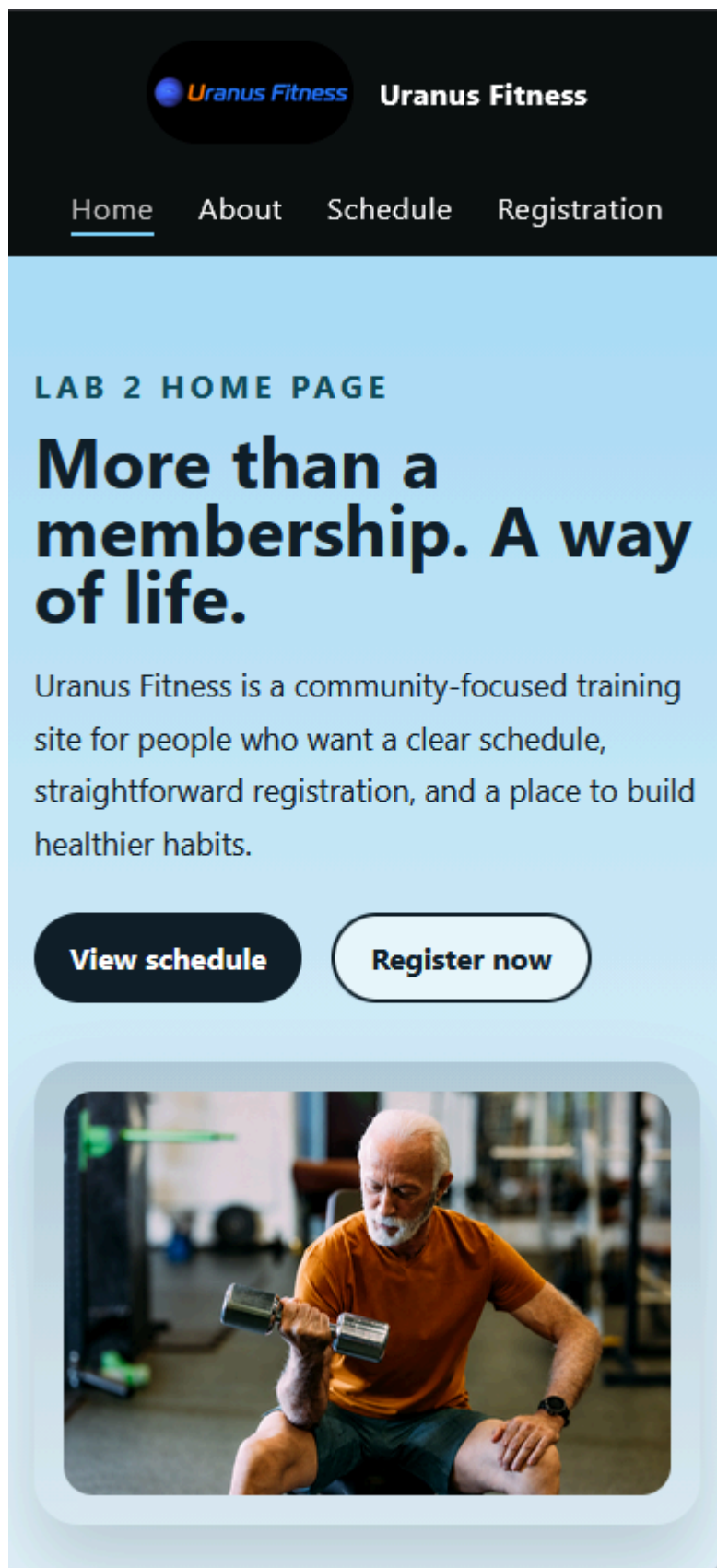


Preferred start date

mm / dd / yyyy



registration.html



WHAT WE OFFER

A clear, well-spaced overview of the workshop experience

Programs and sessions

- Weekly instructor-led workshops covering fundamentals and advanced techniques.
- Open gym sessions for members to practice and socialize.
- Special events and guest instructors throughout the year.

Workshop structure

1. Warm-up and mobility drills (10–15 minutes)
2. Technique instruction and partner drills (20–30 minutes)
3. Conditioning or skill practice (15–25 minutes)

Open training blocks

Unstructured workout time gives members space to practice independently or with a partner.

WEEKLY TABLE

A simple view of the weekly timetable

Uranus Fitness Plans

Plan	Duration	Workout style
Beginner Plan	2 weeks	Machine-assisted exercises
Intermediate Plan	5 weeks	Free weight exercises
Advanced Plan	8 weeks	Compound lifts + diet plans

CONTACT FORM

Registration details

Personal information

Full name

Email address

Phone number

Age range

Select an option



Membership selection

Choose a plan

- ☐ Beginner
- ☐ Intermediate
- ☐ Advanced

7. CSS Validation



CSS Portal

- CSS @Counter Style Preview
- CSS Centering Guide
- CSS Clamp() Calculator
- CSS Code Formatter
- CSS Code Optimizer
- CSS Color Converter
- CSS Color-mix() Calculator
- CSS Cursor Viewer
- CSS Font Preview
- CSS Lengths Viewer
- CSS Logical Properties Converter
- CSS Position Visualizer
- CSS Specificity Calculator
- **CSS Validator**
- Convert Image to Data
- Unused CSS Finder
- HTML Resources >
- Miscellaneous Tools >

REFERENCES

- @ CSS At-Rules >
- ☞ CSS Data Types >
- ƒ CSS Functions >
- 📄 CSS Properties >
- 📄 CSS Pseudo Classes >

Search...

Validation engine: W3C CSS Validator (https://www.w3.org/2011/05/css-validator/)

- @media, @supports, @keyframes, @property, and @container support
- Summary statistics for every stylesheet validated

This validator uses a newly developed CSS validation engine. If you encounter any issues or unexpected results, please report them along with the CSS code used.

If this site has been useful, we'd love your support! Consider buying us a coffee to keep things going strong!

Buy me a coffee

CSS Validator

```
1 /* Base reset and defaults */
2 * { margin: 0; padding: 0; box-sizing: border-box; }
3 body {
4   font-family: system-ui, sans-serif;
5   background: linear-gradient(180deg, #ed7f5 0%, #fff27b 45%, #f7f7fd 100%);
6   color: #10212b;
7 }
8
9 /* Shared accessibility helpers */
10 a { color: inherit; }
11
12 a:focus-visible,
13 button:focus-visible,
14 input:focus-visible,
15 select:focus-visible,
16 textarea:focus-visible {
17   outline: 3px solid #8f4c5c;
18   outline-offset: 3px;
19 }
```

Validate Code Copy Example Code Clear

Results

Congratulations! No errors or warnings found. Your CSS is valid.

<https://www.cssportal.com/css-validator/>

9. Reflection

The original Lab 1 site wasn't built with mobile in mind, so we had to rework several layouts and components to make the pages work on small screens: the header/nav was rebuilt to collapse cleanly, the hero and content areas were converted to responsive CSS Grid patterns with a single media query, forms were refactored into a responsive grid with clear labels and focus styles, and styles were consolidated into one stylesheet for consistency.